

EBR-II INITIAL OPERATION—HIGHLIGHTS

H. O. MONSON*

Work performed under the auspices of the U.S. Atomic Energy Commission

1. Introduction

The Experimental Breeder Reactor-II^(1, 2) has progressed successfully from the preoperational testing phase to the operational phase. This paper describes some of the highlights of that transition and of the operating experience gained. Included are descriptions of the approach to power, selected operational aspects of special interest, fuel element performance, irradiation experiments, and future plans.

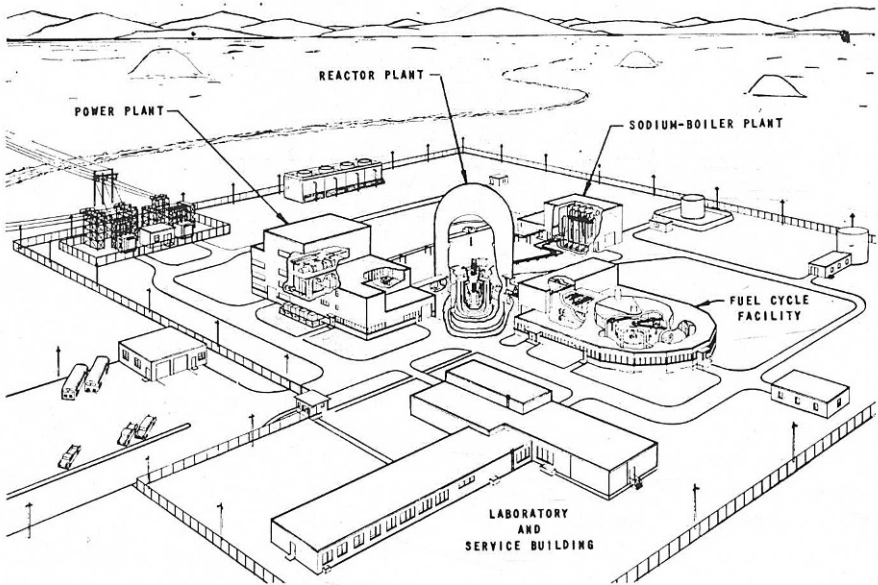


FIG. 1. EBR-II facility.

* Argonne National Laboratory, Argonne, Illinois.

Description of Facility

The EBR-II facility (Fig. 1) is an integral nuclear power plant; i.e. it includes a complete fuel processing and fabrication facility in addition to the reactor, heat transfer systems, and steam-electric plant. The reactor is a

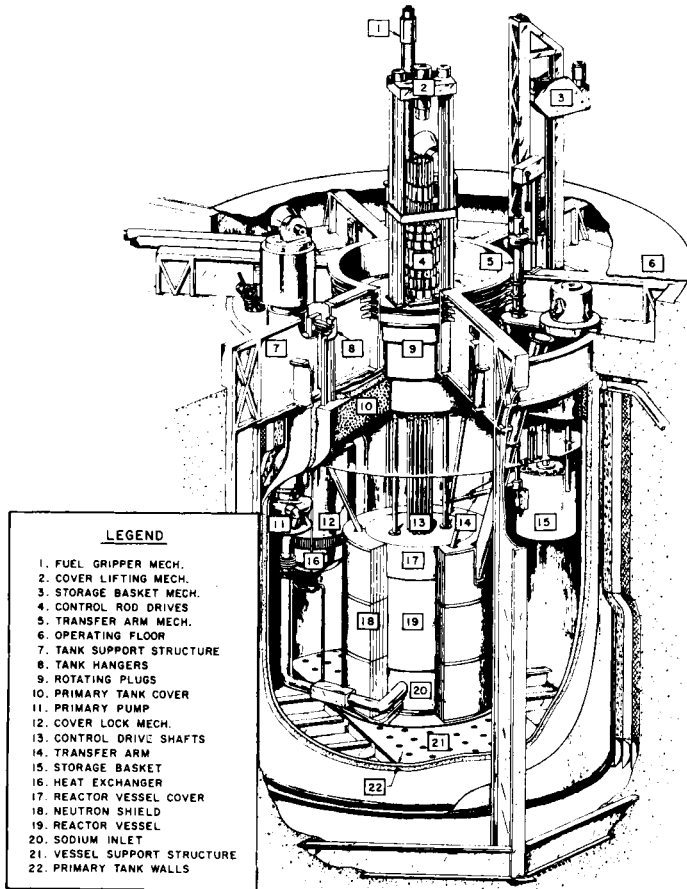


FIG. 2. Primary system arrangement.

fast power breeder, sodium cooled. Maximum power rating is 62,500 kW. Gross electrical output is 20,000 kW. Steam conditions are 1250 lb/in² g (86 atm) and 837°F (447°C).

The reactor and the entire primary system are submerged in sodium contained in a tank of 26 ft (8 m) diameter and 26 ft (8 m) depth. This primary tank has a cover fitted with seventy nozzles through which operating mechanisms, instrument leads, etc., extend. Included among these mechanisms are

the two primary pumps, intermediate heat exchanger, storage basket for spent fuel, fuel transfer arm, and immersion heaters. Two large rotating plugs, one eccentrically located in the other, are employed in the centre of the cover. These carry the fuel gripper, control rod drives, and reactor cover lifting mechanism. The entire tank weighs about 900 tons (including 320 tons of

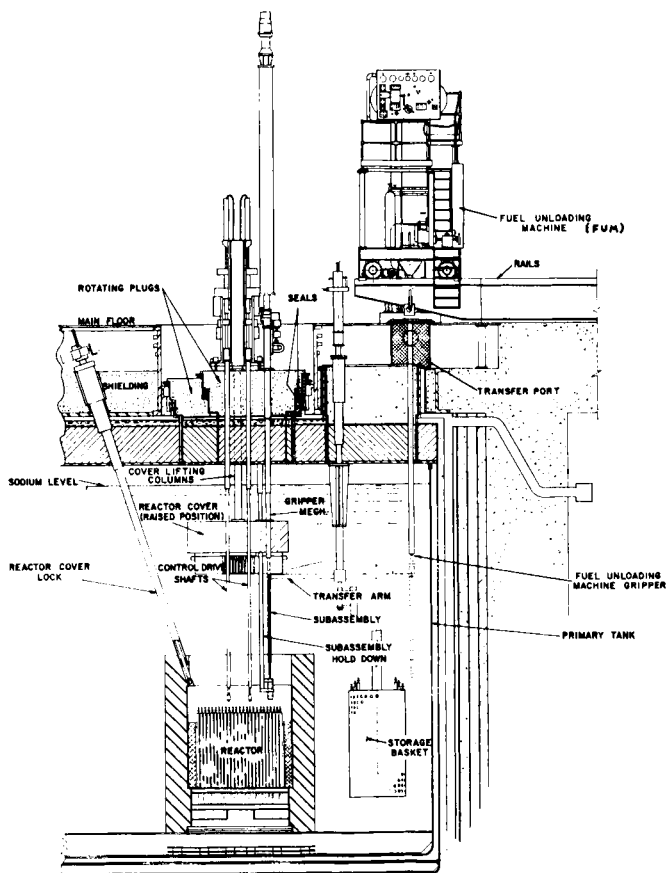


FIG. 3. Principal components of fuel handling system.

sodium) and is suspended from an overhead support structure by six hangers equipped with rollers at their top ends. A pictorial representation of the overall arrangement appears in Fig. 2. The general fuel handling scheme is depicted in Fig. 3.

The reactor core is approximately hexagonal in cross-section, with length to diameter ratio of about 0.8. The fuel alloy is U-5% Fissium. At full power, maximum power density is 1230 kW/litre of core volume. Reactor inlet coolant temperature is 700°F (371°C); outlet temperature is 883°F (473°C).

Breeding blankets of depleted uranium surround the core. Reactor control is effected by movement of fuel into and out of the core. A total of twelve peripherally located control rods are employed, of which any one may be used for regulation. A vertical section of the reactor is shown in Fig. 4.

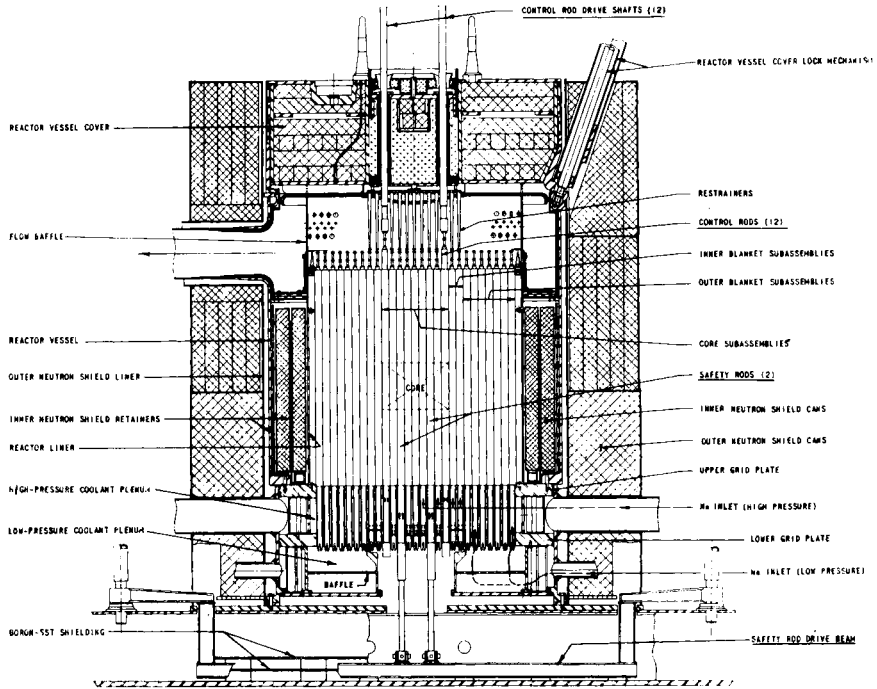


FIG. 4. Reactor vertical section.

2. The Approach to Power

Figures 5 and 6 summarize the power levels and cumulative thermal energy attained throughout the approach to power. Notation is made on Fig. 5 relative to major events which occurred during periods of reactor shutdown. Supplementary information is given below.

The approach to power started on July 16, 1964. Power was raised in steps from 0.5 to 1, 2, 3, 4, and 5 MW and then to 10, 20, and 30 MW. Power and flow coefficients of reactivity were determined and transfer function measurements made at several of these power levels.

A demonstration run at 30 MW was conducted from August 13 to 17 with the turbine generator in operation for the first time and producing 7.5 MW(e). The reactor then was shut down for the first removal of fuel for inspection.

In October, reactor power was raised to 37.5 MW with the generator

producing 11.8 MW(e) and delivering 8.8 MW(e) net to the National Reactor Testing Station electrical system.

At the end of the above run, two control rod drives were found stuck. A 4-month investigative and corrective programme followed. The trouble was found to have stemmed from failure of a ball bushing in the oscillator

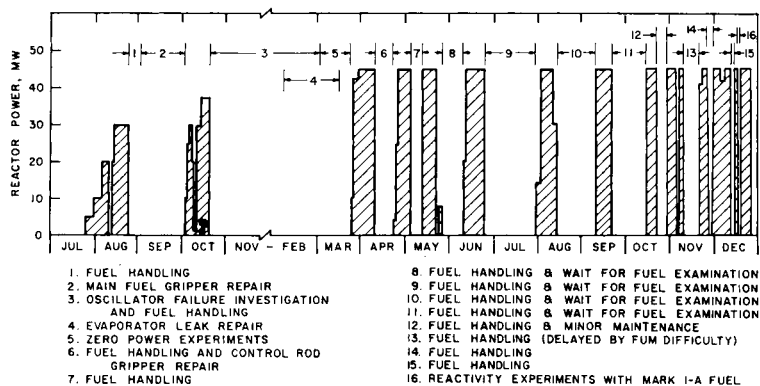


FIG. 5. Power operating history and major events, 1964-5.

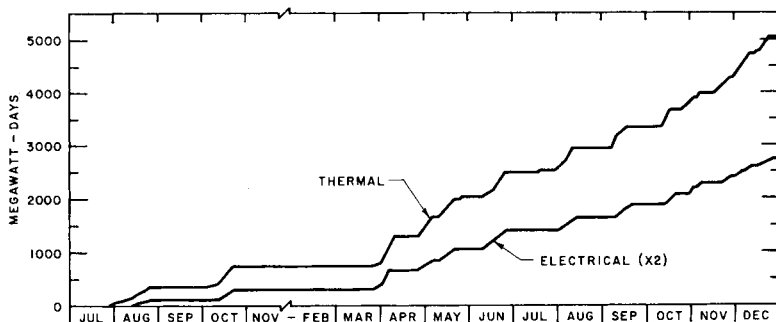


FIG. 6. Integrated power production, 1964-5.

assembly temporarily installed in the reactor for transfer function measurements. While the oscillator was being removed from the system, a pinhole leak was discovered in a tube-to-tubesheet weld in one of the eight evaporators of the steam generator. About a month was required for repair and leak testing of this unit—accomplished without removing the unit from the system.

In March 1965 reactor power was increased to 45 MW, and the generator output to 15 MW(e). Operation was continued at that power level thereafter, with shutdowns essentially only for inspection of fuel and conduct of physics experiments.

It should be noted that, in general, the frequency and duration of reactor shutdowns during the approach to power were dictated by fuel surveillance considerations. A very conservative fuel inspection programme was adhered to because of the uniqueness of the core design and the high thermal performance demanded of it. One result of this was that the achievable plant factor was reduced from what it otherwise could have been. Particularly was this true of the period May–October 1965, when as each burnup increment accrued, the reactor was shut down and prohibited from starting up again until examination of fuel from the preceding run was completed. The efficacy of this procedure, however, is attested by the fact that an achievable burnup value has been demonstrated which is in excess of the minimum design value, and the reactor has been operated to date without so much as a single fuel element failure.

In summary, most of the principal facility design objectives were realized in the course of the approach to power. Reactor performance, both thermal and neutronic, was determined to be as predicted. Extensive experience in operation and maintenance of large sodium systems and components in a nuclear power plant environment was obtained. And in-core experimental irradiations of promising fast reactor fuels—including high burnup capsules of oxide, carbide, and metal fuels—were initiated.

3. Selected Aspects of Operation

Reactor and Primary System

Performance of the reactor and primary system has been excellent. The reactor is highly stable and readily controllable. All core power densities, coolant flow rates, and subassembly outlet coolant temperatures are as predicted (one out of the total of twenty core subassembly outlet thermocouples reads higher than calculated, but this couple is believed to be incorrectly positioned). The reactor inlet coolant temperature is easily maintained at the desired level with precision (in the submerged concept used, mixing of the large mass of primary bulk sodium effectively minimizes reflection of steam or secondary system load changes). Primary and secondary system thermal power measurements agree within a few per cent.

Mechanical stability of core components is good. Examination of the entire first core loading after removal from the reactor has shown no evidence of mechanical weakness of any kind. No wear or fretting marks or other indication of vibration, either of fuel elements or of subassemblies, has appeared. Similarly, no suggestion of coolant flow blockage, surface deposition, or any sort of deleterious effect from particulate or other coolant contamination has arisen. No gas entrainment in the coolant has been discerned. No sticking of subassemblies in the reactor vessel grid has occurred.

Because no fuel element failures have occurred, virtually no fission product buildup outside the reactor has occurred.

Neutron and gamma shielding provisions have proved more than adequate. Even in the complex primary tank cover area with its numerous nozzles and mechanisms, no shielding additional to that provided in the original design has become necessary. Effectiveness of the primary neutron shielding immediately surrounding the reactor vessel is illustrated by the fact that after more than 5000 MWd of reactor operation, stainless steel heater assemblies removed from the primary tank generally exhibited surface activity levels of less than 30 mr/h.

Maintenance of the reactor and primary system has proved entirely feasible and practicable. (The oscillator failure caused substantial delay; however, this temporarily installed experimental unit was not considered a part of the regular primary system.) Repair or replacement of numerous components was accomplished without difficulty. Included among those partially or completely removed from the system and then reinserted were several control rod drive mechanisms, nuclear instrument assemblies, primary tank heaters, auxiliary gripper plug, and reactor vessel cover lock mechanisms. A number of minor difficulties, and one mishap which delayed reactor operation a month, occurred with fuel handling mechanisms and were surmounted. These are described in a later section.

Secondary and Steam Systems

Operating experience with the secondary sodium system has been very good. No significant troubles have developed. Intermediate heat exchanger performance is as predicted, both thermally and hydraulically. Sodium activity induced by passage through the heat exchanger (located in the primary tank) is negligible—only a few mr/h can be observed at system pipe surfaces. The 60 c/s induction heating employed on the entire secondary has worked well, although minor improvements in its control system are now planned. Draining and refilling of the system, usually to facilitate repair or maintenance of steam system components, has been accomplished repeatedly and easily.

Steam system performance to date has been fair. As seems normal with nuclear plants, most troubles have developed with conventional, non-nuclear components. Only indicative difficulties will be noted here. The steam turbine driving the boiler feedwater pump failed early in operation, requiring replacement of the turbine wheel. The positive displacement start-up feedwater pump malfunctioned repeatedly and is to be replaced. Steam drum and main steam header safety valves frequently relieved at unacceptably low pressures or failed to reseal properly, necessitating numerous checks and adjustments. Steam bypass valves often stuck full open or controlled erratically; several

new sets of trim and a new actuator were required. Bellows failures occurred in conventional expansion joints in the steam piping system, requiring re-design.

Experience with the sodium-to-water steam generator has been better than expected. No sodium-to-water leaks have occurred. Pressure drops are lower than anticipated, and dissipation of full 62.5 MW reactor power is possible with only the two superheaters presently installed (four were called for in the design). Although detailed analyses have not been made, heat transfer coefficients appear to be essentially as predicted. The only difficulty encountered to date has been the single occurrence of a pinhole leak of water to atmosphere as mentioned earlier, and this was repaired with little effort. At the 45 MW level employed to date, steam conditions are 1250 lb/in² g (86 atm) and 795°F (424°C).

Fuel Handling

The large amount of fuel handling accomplished has firmly established the feasibility of the under-plug, frequent operation⁽³⁾ fuel handling scheme for sodium cooled reactors. Inclusive of the pre-operational testing period, more than 1200 fuel transfers have been made. Since start of the approach to power, 167 core type subassemblies (about 2½ cores) and numerous blanket subassemblies have been replaced. About 50 of the core subassemblies removed had reached burnup of 1 at.% or more. Many high burnup subassemblies have been removed from the primary tank to the reprocessing facility within only 9–15 days after reactor shutdown—with decay heat generation of the order of 2 kW/litre of fuel.

Fuel handling experience has not been completely without difficulty, however. During pre-operational testing, numerous equipment and procedural problems came to light which required design changes and operating refinements. After start of power operation, minor difficulties continued to crop up in several fuel handling areas. These have caused little delay in operation (with two exceptions as noted later), but have been a source of annoyance.

Difficulty frequently was encountered in rotating the large rotating plug to position the fuel gripper. This phenomenon, a tendency towards sticking of the plug during initial operation, has persisted to the present time and despite investigation is still unexplained. It is thought to be related either to operation of the plug freeze seal or to accumulation of sodium and sodium oxide in the annular gap between plug and structure. Once the plug has been started moving, by pushing, it starts and rotates freely throughout the remainder of the fuel handling period.

The fuel unloading machine (FUM) is subject to sticking problems, also. The sensing device which indicates whether a subassembly is held in the machine's gripper jaws works with difficulty. In addition, abnormal force

sometimes is required to raise or lower the gripper through the transfer port of the primary tank. Both of these troubles are related to the presence of sodium or sodium oxide on the gripper mechanism and necessitate its occasional removal and cleanup (and cleanup of the port). The process normally requires about 2 days' effort, and can be accomplished during reactor operation. In one instance (event 13 in Fig. 5), however, more than a week's delay resulted from this source. The transfer arm sensing device has evidenced the same sort of sticking difficulty, but this has not been serious enough to require removal of the arm.

The control rod drive grippers, which are required to disengage from the rods prior to fuel handling, also have given trouble. On one occasion, two grippers malfunctioned so that proper release of the rods from the drives could not be obtained. Both drives were removed and inspected. On one, a small cam necessary for jaw operation had broken loose (remaining inside the gripper). On the other, only burnishing of cams and slight galling of the gripper sleeve were evident. Minor modifications were easily made on two spare units and these units then installed. This difficulty has not recurred since. On another occasion, a drive gripper had to be replaced because of sensing device sticking. It is believed this sticking was caused by failure to remove all water from the drive internals subsequent to a cleaning operation required when this and another rod drive had to be removed during investigation of the oscillator failure incident. The replacement gripper (new) is now not working properly and is to be removed for inspection.

Another delay in reactor operation attributable to the fuel handling system (event 2 in Fig. 5) occurred in September 1964. During fuel handling operations, the main fuel gripper struck the subassembly holddown device inside the primary tank and was damaged. Investigation showed the interference had resulted from thermal distortion of the gripper shaft caused by having maintained the shaft unused in the up, cool, position for an excessive length of time. A month was required to remove, repair, and reinstall the unit. A change in operating procedure has precluded recurrence of the incident.

Of greatest significance is the fact that the time required for fuel handling is short and has been progressively becoming shorter. At the end of a reactor run in December 1965, fuel handling (event 15 in Fig. 5) was accomplished in 31 hours, from start of reactor shutdown from 45 MW to completion of fuel handling. In that period, ten subassemblies, or about one-seventh of the core, were removed from the reactor and replaced. On January 10, 1966, total time from reactor shutdown from 45 MW to completion of fuel handling (replacement of one core subassembly) was 13 hours.

Physics

Physics aspects of the reactor are well understood. No anomalies of significance have developed. Values for selected parameters, comparing

predicted values with measured values, are shown in Table 1. It is interesting to note that the worth of control rods located on the core flats has changed

TABLE 1. SOME PREDICTED AND MEASURED EBR-II NUCLEAR PARAMETERS

Parameter	Predicted after dry critical	Measured
Critical mass, kg U ²³⁵	180	181.2*
Reactivity worth, % $\Delta k/k$		
Control rod (average of 12)	0.34	0.345
Control rods (total of 12, banked)	—	4.4
Safety rods (total of 2)	1.3	1.2
Isothermal temp. coef., $[\Delta k/k/^\circ\text{C}] \times 10^{-5}$		
Structure (including fuel)	-2.2	—
Coolant	-1.8	—
Total	-4.0	-4.4*
Power calibration, counts/w-sec	500	390
Subassembly substitution worth, % $\Delta k/k$		
At core centre	1.26†	1.2†
In row 5 (radius = 21 cm)	0.64†	0.56†
In row 6 (radius = 27 cm)	—	0.35†

* At 600°F (315°C).

† Mark I enriched subassembly versus natural uranium subassembly of otherwise identical composition.

TABLE 2. CALCULATED POWER COEFFICIENTS OF REACTIVITY, $[\Delta k/k/\text{MW}] \times 10^{-5}$

Description	Power range (MW)	Without CR drive effect*	With CR drive effect	
			50% inserted†	100% inserted†
Without bowing or radial expansion	0-62.5	-3.2	-4.9	-3.7
Without bowing, with radialexpansion	0-62.5	-4.3	-6.0	-4.8
With most probable bowing and expansion	0-25	-3.5	-5.2	-4.0
	25-34	+1.0	-0.7	+0.5
	34-62.5	-4.0	-5.7	-4.5
With pessimistic bowing and expansion	0-20	-3.4	-5.1	-3.9
	20-56	+0.9	-0.8	+0.4
	56-62.5	-4.3	-6.0	-4.8

* Effect on reactivity resulting from change in control rod position caused by thermal expansion of rod drive shaft.

† Control rod insertion.

somewhat as operation has progressed and core configuration and composition have changed (because of experimental irradiation subassembly additions). Average worth has decreased from 0.382% $\Delta k/k$ per rod initially to

about $0.365\% \Delta k/k$ at present. Loss of reactivity attributable to fuel burn-up and plutonium production has been determined to be $-3.4 \times 10^{-6} (\Delta k/k)/\text{MWd}$.

The predicted power coefficients of reactivity at full coolant flow are shown in Table 2. Numerous measurements throughout the approach to power indicated the reactivity decrement for power level change from zero to 45 MW to be in the range 66–83 inhours under all conditions. The average was 77 inhours, corresponding to an overall average power coefficient of $-4.1 \times 10^{-5} (\Delta k/k)/\text{MW}$. The differential coefficient varies with power level, approximately as follows: from 0 to 20 MW, -4.3×10^{-5} ; from 20 to 30 MW, -3.5×10^{-5} ; and, from 30 to 45 MW, $-4.1 \times 10^{-5} (\Delta k/k)/\text{MW}$ (all measured with 80–90% control rod insertion). This variation probably is caused by bowing, and was anticipated. The measured coefficients agree reasonably well with those calculated for the most probable bowing configuration shown in the table.

During early operation (July–October 1964), two independent sets of transfer function measurements were conducted at power levels to 37.5 MW using a reciprocating oscillator rod and frequencies from 0.005 to 1.7 c/s. For one set, a wave analyser operating on the null-balance principle, was used. For the other, a cross correlation Fourier analysis of power response was made using an analogue computer. Subsequent to failure and removal of the oscillator mechanism, the stability monitoring programme has consisted of: trapezoidal oscillation using a regular control rod; rod drop measurements; and continued power coefficient measurements. With the trapezoidal technique, the peak-to-peak amplitudes of the power response recordings for both operating power and zero power are analysed to reflect long time constant (10–100 sec) behaviour. With the rod drop technique, using a special steel, dummy control rod of 17 inhours' worth, the prompt behaviour is determined. Analysis of power response from repeated drops suggests that the prompt feedback might be generally characterized by two terms: one with a time constant of about 0.3 sec and amounting to some 40% of the total power coefficient; and one of about 0.5 sec time constant, amounting to 30% of the total. All monitoring results to date indicate that the reactor is highly stable and that no significant change in prompt or delayed components of the power coefficient has occurred with increasing burnup or changes in core composition or configuration.

Instrumentation

In general, instrumentation has been satisfactory. Performance of some elements of the originally supplied instrumentation in the primary system, however, has been poor. During preoperational testing and the approach to power, the following primary system instruments failed: nine of twelve

resistance thermometers (platinum element, magnesium oxide insulation, stainless sheath); thirteen of seventeen pressure sensors (diaphragm actuation, NaK-filled capillary leads); and two of five flowmeters (modified venturi type). All five primary flowmeters of the electromagnetic type and most thermocouples in the system, on the other hand, have consistently performed well. Because of redundancy of sensors and extensive calibrations

TABLE 3. UNSCHEDULED REACTOR SCRAMS, 1965

Month	Number of scrams	Initial power (MW)	Cause
Mar.	3	45, 25, 6	Instrument fault
	1	37	Operator error
	2	45, 37	Pri pump CW Pr low
Apr.	1	4	Instrument fault
May	2	45, 6	Instrument fault
	1	10	Lost 138 kV power
June	6	45(4), 20, 8	Instrument fault
	1	20	Adjustment of trip
July	1	15	Instrument fault
	2	15(2)	False FERD signal
Aug.	3	45(2), 15	Lost 138 kV power
	1	30	False FERD signal
Sept.	3	45(3)	Instrument fault
	1	15	Pri pump CW Pr low
	2	45(2)	False FERD signal
Oct.	3	22, 20, 10	Instrument fault
	1	45	Operator error
Nov.	3	45(2), 20	Instrument fault
	1	45	False FERD signal
Dec.	5	45(2), 10, 5	Instrument fault
	1	45, 42	Operator error
	2	45, 18	False FERD signal

made prior to the sensor failures described, thermal and hydraulic monitoring of the primary system remains satisfactory. Nuclear instrumentation performance has been good throughout operation.

Minor faults in instrumentation have produced a number of unscheduled reactor scrams. The most frequent cause has been noise. The fuel element rupture detection (FERD) system, still undergoing development, is particularly vulnerable in this respect. All unscheduled scrams (from all causes) experienced during power operation in 1965 are listed in Table 3.

Pump Performance

During preoperational testing, both of the primary pumps (centrifugal type, 5000 gpm each) failed soon after start-up. The cause was rubbing of the

pump shaft at the lower labyrinth seal. This seal had been provided with small clearance to minimize migration of sodium vapour. Before start of the approach to power, the pumps were reworked to incorporate shafts and seals of improved design. Since then, both pumps have accumulated about 7000 hours' operating time, most of this being at full reactor flow. In July 1965 one pump started sticking as a result of sodium or sodium oxide accumulation above the seal. The sticking was easily overcome by vertical exercising of the shaft. An increase in argon backflow along the shaft has precluded recurrence since. Aside from this minor difficulty, operation has been completely satisfactory.

The secondary system pump (electromagnetic induction type, 6500 gpm) also was repaired during preoperational testing. It earlier had developed a leak owing to fatigue cracks caused by duct wall vibration at high flow. As insurance against further cracking, operation of the repaired pump presently is limited to 85% of full flow. Since start of the approach to power, the pump has accumulated approximately 10,000 hours' operating time and has performed completely satisfactorily.

The secondary pump control system enables stable flow at rates down to a small fraction of 1%. It also enables reverse thrust. These characteristics are needed during plant standby when extremely low secondary flow (frequently lower than that effected by natural convection alone) is required. Despite a small control dead-band in the zero thrust region, operation of this system has been excellent.

Coolant Purity

Oxide concentration in the primary system sodium has remained low since the filling of the primary tank in February 1963. Plugging temperature

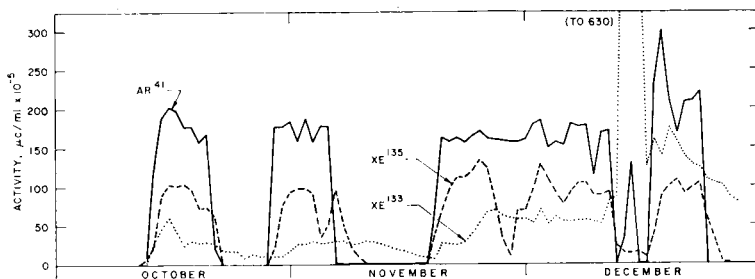


FIG. 7. Primary system cover gas activity (typical, 1965).

is kept below 250°F (121°C) by cold trapping only 2 or 3 days per month. Since the start of power operation, all available evidence (examination of fuel elements) indicates essentially complete absence of particulate or other gross contamination in the coolant. Primary cover gas normally contains 10 ppm

or less hydrogen and 1–2% nitrogen; some activity levels are shown in Fig. 7.

The secondary system, because of presence of piping corrosion products, initially required several dumpings and prolonged cold trapping to reduce oxide content. Since that time, plugging temperature normally has been kept below 300°F (149°C) by intermittent cold trapping. Actual plugging temperature variation and cold trapping operation for a typical quarter are illustrated in Fig. 8. Secondary cover gas normally contains 10 ppm or less hydrogen and 0.1–0.3% nitrogen.

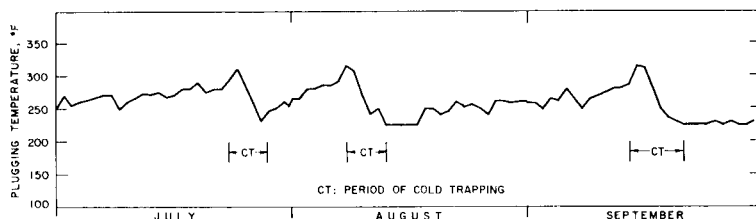


FIG. 8. Secondary system plugging temperature (typical, 1965).

Fuel Element Rupture Detection

Two systems are provided for fuel element rupture detection: a delayed-neutron monitor operating on a side stream of reactor outlet sodium; and a charged-wire fission gas monitor operating on the primary system cover gas. Initial calibrations for both systems have been obtained by alternately moving into and out of the reactor core during power operation a special control rod incorporating a bare Mark I fuel alloy pin. Power levels of zero and 10 MW and flow rates of 50% and 70% were employed. Results clearly indicated that both systems were capable of detecting fission product activity associated with a single unclad fuel pin. As expected, time response of FERD (15–20 sec) is very much faster than for the charged-wire system (hours). The FERD normal background neutron level is approximately 0.5 c/s. MW (uranium content of the sodium is less than 0.5 ppm); normal background sodium gamma level at 45 MW is 5–10 c/s (activity of the sodium is about 0.6 curies/litre). With the charged-wire system, which monitors Cs^{138} , Rb^{88} , and Rb^{89} , the signal to noise ratio obtainable is somewhat superior to that for FERD. Additional calibration work at higher power levels is planned for both systems.

Fuel Cycle Facility

A complete fuel reprocessing plant, or fuel cycle facility, is provided at EBR-II. Its principal functions are to receive spent core type subassemblies from the reactor and clean them of sodium; disassemble the subassemblies

and strip their fuel elements to obtain the fuel alloy pins; pyrochemically process the pins (by melt refining) to remove the majority of the fission products; reconstitute the alloy and recast it into new fuel pins; and, using the new pins, assemble and test new fuel elements and subassemblies for insertion in the reactor. All of these functions necessarily are accomplished remotely.

The fuel cycle facility has been placed into operation essentially simultaneously with the reactor facility. No major problems have developed, although numerous minor difficulties have been encountered requiring equipment design changes, procedure revisions, and extensive maintenance. To date, 2441 fuel elements and fifty-one subassemblies employing unirradiated alloy have been assembled. Some 7000 irradiated elements (eighty-five subassemblies) with burnups up to 1.2 at. % have been reprocessed. More than 3300 new elements (and forty-three subassemblies) have been fabricated, partially or entirely, of the reprocessed fuel. Currently, the facility is reprocessing about ten subassemblies per month. This rate is expected to increase in the future.

The reactor presently contains sixty-seven subassemblies fabricated in the fuel cycle facility. Of these, thirty-four contain second-cycle fuel alloy (irradiated and reprocessed once).

4. Performance of Mark I Fuel Element

An extensive surveillance programme was conducted to enable assessment of performance of the Mark I fuel element (Fig. 9) used in the first core

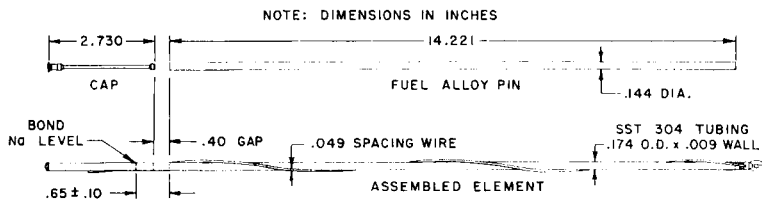


FIG. 9. Mark I fuel element.

loading. At frequent intervals, as burnup of fuel progressed, the reactor was shut down and one or more selected subassemblies were removed for disassembly and examination. Normally, all ninety-one fuel elements of a subassembly were visually inspected. Some seventeen to twenty-one elements from each were given diameter, bond integrity, and bond sodium height checks. About ten of these then were stripped for fuel alloy pin examination. The most useful determination was that of the sodium height. As burnup proceeds, the fuel alloy grows in volume and displaces bond sodium. As the sodium height increases, the gas volume at the top of the element grows

smaller and the gas pressure increases. Because maximum achievable burnup is limited by the allowable internal pressure, sodium height is a valuable indicator as to whether additional burnup might be safely accrued. Also, the change in height is quantitatively indicative of the amount of growth experienced by the fuel alloy. Figure 10 shows approximate fuel growth versus maximum local burnup as calculated from measured changes in height. Circular data points represent subassembly averages; square data points depict the maximum-growth element in each subassembly. The maximum local burnup indicated is approximately 1.14 times the average burnup of the highest burnup element in the subassembly. Almost all burnup was

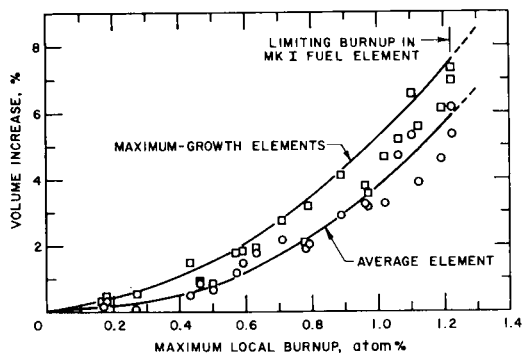


FIG. 10. Fuel alloy (U-5% Fs) growth as a function of burnup.

accrued at reactor operating conditions of 45 MW and full flow, for which fuel alloy temperatures ranged from about 720°F (382°C) at inlet ends of all elements to about 1050°F (566°C) at the highest temperature portions of the hottest elements.

In brief summary of the fuel surveillance results, Mark I fuel element performance appears excellent. All fuel elements have been found in good condition, with no discernible change in external diameter. Of the more than 2000 regular Mark I fuel elements removed for inspection, not one has been found defective. The minimum design burnup level (1.0 at.%, local) has been exceeded; a practicable limit of 1.22 at.% has been established and is in use.

A modified version of the Mark I fuel element, Mark IA, has been placed in production. It embodies two changes: fuel alloy pin length is shortened from 14.221 in. (36.121 cm) to 13.50 in. (34.29 cm) (with correspondingly enlarged gas space at the top of the element); and enrichment is increased from 48.4% to 52.5%. The former change should increase substantially the maximum burnup achievable. The latter, by increasing reactivity, should enable accommodation of a larger number of irradiation experiments in the reactor. Loading of the new elements has already started; by mid-summer 1966 the entire driver core is expected to be composed of Mark IA elements.

5. Irradiation Experiments

To date, a total of 105 irradiation experiments, including oxide, carbide, and metal fuels, have been inserted in the reactor. Most are located in the

TABLE 4. IRRADIATION EXPERIMENTS IN CORE AND INNER BLANKET

Reactor zone	Sub-assembly	Numbers of experiments by type				Total in sub.	Total in zone
		Oxide fuel	Carbide fuel	Metal fuel	Structural material		
Core	*XG01	6	—	—	4	(10)	—
Core	*XA01	—	—	19	—	(19)	—
Core	XG05	9	3	2	5	(19)	—
Core	XG06	12	—	2	5	(19)	—
Core	XA07	—	—	16	3	(19)	—
Core	XA08	—	8	—	11	(19)	—
		(27)	(11)	(39)	(28)		(105)
IB	XG02	1	—	—	—	(1)	—
IB	XG03	2	—	—	—	(2)	—
IB	XG04	2	—	—	—	(2)	—
		(5)					(5)
							(110)

* Irradiation completed.

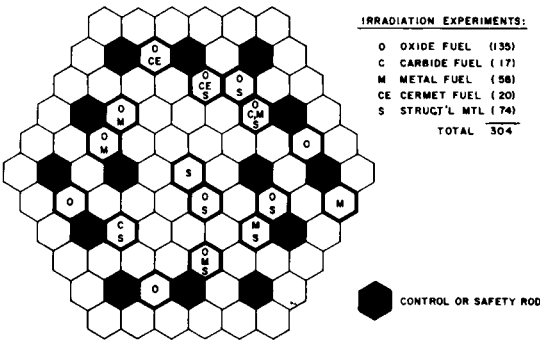


FIG. 11. Possible future loading of experimental irradiation subassemblies in core (summer 1966).

core, with a small number in the inner blanket as indicated in Table 4. In addition, 176 materials surveillance experiments (approximately 2000 specimens) have been placed in the outer blanket.

It is intended rapidly to expand the number of experiments in the core. Figure 11 illustrates a possible future loading. Whether the exact number of experiments indicated (304) actually can be accommodated will depend upon reactivity considerations. Also, the indicated fuel type distribution may well be altered, very likely to emphasize metals less and carbides more.

6. Future Plans

In the future, primary emphasis will be placed upon utilization of EBR-II as a fast flux irradiation test facility. To aid in this objective a number of improvements are being made or are under consideration. New, special types of experimental irradiation subassemblies will be put into service. These will permit use of longer test specimens and higher test section coolant temperatures (now limited to, roughly, 1000°F (538°C)). Changes in inner and outer blanket subassembly designs may be made to eliminate uranium (substituting nickel or stainless steel) and reduce coolant flow rates. These changes would flatten neutron flux, enable accommodation of additional experiments (by increasing reactivity), and free additional coolant flow for use in the core. Further modifications to the present core fuel element design will be made to increase maximum achievable burnup. This will reduce the frequency of shutdowns for refuelling and improve plant factor. A new, full-capacity secondary system pump is expected to become available about the end of the year; reactor power then will be started upwards towards 62.5 MW.

References

1. L. J. KOCH and H. O. MONSON *et al.*, *Hazard Summary Report, Experimental Breeder Reactor II (EBR-II)*, ANL-5719 (May 1957). Also see Addendum (June 1962).
2. L. J. KOCH and H. O. MONSON *et al.*, Construction design of EBR-II: an integrated unmoderated nuclear power plant, *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy, Geneva, 1958*, Vol. 9, p. 323.
3. L. J. KOCH, Fast reactor fuel handling, *ANS National Topical Meeting, Detroit, Michigan, ANS-100* (April 1965), p. 333.